

Mark O'Dowd

SENIOR FULL STACK ENGINEER

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SUMMARY

Senior Full Stack Engineer with over 7 years of experience building and architecting scalable, cloud-native applications. Proven track record of bridging product vision with high-performance production while reducing infrastructure overhead through automated, cost-optimized deployments. AWS AI Certified, specialising in leveraging GenAI models and cloud ML services to build secure, enterprise-grade intelligent features

TECHNOLOGY STACK

Languages: TypeScript, Python, Rust
Backend: Django, Node.js, Express, Axum
Infrastructure: AWS, GitHub Actions, CI/CD

Frontend: React, Next.js, Tailwind CSS
Databases: PostgreSQL, SQLite, DynamoDB, MongoDB
Tools: Linux, Bash, Docker, Neovim, Tmux

EXPERIENCE

FULL STACK CLOUD ENGINEER (FREELANCE/CONTRACT) - Emerald Software - Mar 2025 to Present

www.irishtrad.ie - Architected and deployed a high-performance, full-stack educational platform for traditional Irish music instruments

- **High-Performance Architecture:** Built a dual-engine architecture utilising a server-side rendered Next.js frontend for optimal performance and a low-latency, high-throughput Rust (Axum) backend REST API
- **Subscription & Payment Processing:** Engineered a secure billing infrastructure featuring premium subscription tiers and token-based payment flows, seamlessly integrated with Stripe
- **Cloud Infrastructure & Media:** Designed and deployed a robust AWS cloud architecture leveraging ECS & Fargate for deployment, S3 for media asset hosting, RDS for database management, and SES for reliable transaction email delivery

www.fightorflight.studio - Developed a bespoke, media focused portfolio for a Manchester-based creative studio

- **Headless CMS Integration:** Engineered a highly customizable portfolio platform leveraging Contentful CMS, empowering the client to independently manage and update media assets without developer intervention
- **Media Optimization:** Implemented adaptive bitrate streaming for high-resolution video assets, drastically reducing initial page load times and improving mobile performance
- **CI/CD & Infrastructure:** Automated deployment pipelines for staging and production environments to ensure seamless delivery, and secured domain email deliverability by configuring SPF, DKIM, and DMARC protocols through Cloudflare

www.jujitsu.ie - Developed a modern, mobile-first web application to digitize the World Ju-Jitsu Federation Ireland syllabus

- **Zero-Maintenance Hosting:** Designed a serverless, highly cost-efficient hosting architecture utilizing an AWS S3 static bucket, virtually eliminating ongoing infrastructure overhead.
- **Performance Optimization:** Optimized asset delivery by implementing modern WebP image compression, significantly reducing page weight and boosting load speeds.
- **Mobile-First Design:** Engineered a responsive, lightweight, mobile-first user interface to ensure student syllabus materials are highly accessible on any device during training.

SOFTWARE ENGINEER - Dawson Andrews - May 2024 to Mar 2025 - Dawson Andrews specialises in product design, strategy, and software development

- Rapidly adopted the Vue/Nuxt ecosystem to lead the end to end development of an enterprise grade web application ensuring a seamless transition from design to AWS backed production

SOFTWARE ENGINEER - Hurree - Jan 2023 to Mar 2024 - Hurree is a cross platform analytics dashboard that enables teams to unify data to gain one high level overview

- Engineered user friendly analytics features using React and Django, unifying unique data sources into high-performance, real-time dashboards for cross-platform business intelligence while championing code quality and system reliability by leading technical design discussions and conducting rigorous code reviews

SOFTWARE ENGINEER - Micro Focus - Apr 2022 to Nov 2022 - Micro Focus is one of the world's largest enterprise software providers delivering mission critical technology

- Spearheaded the integration of React into a mission critical legacy Java ecosystem creating a scalable i18n framework that enabled global localisation
- Provided mentorship and guidance to junior engineer on my team teaching how to do pull requests

WEB ENGINEER - VML Technology - Nov 2019 to Mar 2022 - VML Technology pioneers new levels of video maximizing engagement through real time convergence of data

- Delivered core web platform features using React and Django, achieving 1:1 parity with mobile application for real time video engagement

TECHNICAL LEADERSHIP & COMMUNITY

- Star Mentor: Zero to Mastery** is an online academy providing video courses on a wide range of programming topics with a Discord community of over 500,000 members. After 7 years of dedication towards learning and giving back to the community through mentorship of juniors I am now 1 of 20 people in the community that has been recognised and granted the title of Star Mentor
- Tech Talks:** Passionate about education I have spoken at tech events within the data engineering community covering topics such as data manipulation with Apache Spark and how to train your own model with Spark ML
- Meetups:** Engaging and networking within the various tech communities is important in order to stay on top of current trends and as such I have attended ~30 different events within the Dublin area: [Meetup Profile](#)
- Technical Blog:** I'm an avid book reader and have recently started doing a blog reviewing each book. The blog is available at [Personal Blog](#)
- Open Source Contributions:** As a Linux user I believe in giving back to open source software when I identify an issue or some way in which I can contribute. Some examples of contributions:
 - React Flow:** I noticed issues during development with the TypeScript typings in the library so provided a detailed explanation and discussion to fix: <https://github.com/xyflow/web/issues/479>
 - TanStack Table:** When the Table project was early in development while reading the docs I identified issues and provided a detailed video which has since been completed: <https://github.com/TanStack/table/issues/5193>
 - Leptos Book:** Leptos is one of the leading frameworks in the Rust web ecosystem. While reading the documentation there was grammatical issues. I provided a fix: <https://github.com/leptos-rs/book/pull/223>

CERTIFICATIONS

- AWS Certified Solutions Architect - **SAA-C03** - Amazon Web Services - [Verify](#)
- AWS Certified Cloud Practitioner - **CLF-C02** - Amazon Web Services - [Verify](#)
- AWS Certified AI Practitioner - **AIF-C01** - Amazon Web Services - [Verify](#)

EDUCATION

Queens University Belfast - BACHELOR OF SCIENCE (Bsc) MUSIC TECHNOLOGY